Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

AI-Powered Ethical Decision Making Simulations with Mistral

Team Name:

Code Crafters

Team Members:

- Pavani Avvaru
- Rasi Supriya
- Yamparala Sravani
- Tejaswini pushadapu

Phase-1: Brainstorming & Ideation

Objective:

- AI-Powered Ethical Decision Making Simulations with Mistral
- The project aims to develop AI-driven simulations that assist individuals and organizations in navigating complex ethical dilemmas. Leveraging Mistral AI's capabilities, these simulations will provide real-time analysis, reasoning, and justifications for ethical decision-making. The goal is to create interactive, scenario-based experiences that enhance ethical awareness, critical thinking, and bias detection in various domains, such as business, healthcare, law, and technology.
- Enhanced Ethical Awareness
- Bias Detection & Mitigation
- Improved Decision-Making
- Interactive & Engaging Learning

Key Points:

1. **Problem Statement: AI-Powered Ethical Decision Making Simulations with Mistral**

2. **Proposed Solution:**

a chartboad is developed to view the multiple solutions in multiple ways to enhance the problem

3. **Target Users:**

*** Corporate & Business Leaders**
*** AI & Technology Developers**
*** Educators & Students**
*** Legal & Compliance Professionals**
*** Healthcare Professionals**
*** Government & Public Policy Makers**
*** General Public & Consumers**

4. **Expected Outcome:**

* Interactive Ethical Decision-Making Simulations
* AI-Driven Reasoning and Explanation Engine
* Bias Detection and Mitigation
* Personalized Learning Paths
* Ethical Decision-Making Reports and Insights

Phase-2: Requirement Analysis

Objective:

- Simulation Scenarios & Ethical Dilemmas
- AI-Powered Decision Support
- User Interaction & Experience
- Personalization & Learning Adaptation
- Collaboration & Multi-User Mode
- Compliance & Transparency

Key Points:

Technical Requirements: AI Model & Integration, Architecture & Backend Frontend & User Interface ,Performance & Scalability,Testing & Monitoring Function

Constraints & Challenges: 1. Monitoring & Auditing AI Decisions 2. User Trust & Acceptance 3. Ethical & Compliance Constraints 4. Computational & Performance Constraints

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Phase-3: Project Design

Objective:

- Create the architecture and user flow.

Key Points:

System Architecture :Dashboard & Reporting, Command & Control Panel ,Real-time Alerts
User Flow: user uses the chartboard for their ethical decision making and can get better decisions to the problem resolution

UI/UX Considerations: Threat Intelligence Database ,Security Logs & SIEM Integration,User & Entity Behavior Analytics (UEBA)

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Stage 1: Project Initialization & Research
- Stage 2: Core Simulation Development
- Stage 3: User Engagement & Learning Path Creation
- Stage 4: Testing & Iteration
- Stage 5: Deployment & exicution

Key Points:

- **Sprint Planning:**
- **Member 1:** Project Initialization & Research,Core Simulation Development
- Member 2:User Engagement & Learning Path Creation
- Member 3: Testing & Iteration
- Member 4: Deployment & exicution

1. **Timeline & Milestones: 2days**

Phase-5: Project Development

Objective:

1. Frontend: React.js, Vue.js (for UI/UX)

2.Backend: Python (FastAPI, Flask) with Mistral AI API for ethical decision-making

3.AI Models: Mistral AI (Natural Language Understanding), OpenAI API for contextual ethics

4.Data Processing: PostgreSQL, MongoDB (to store decisions, logs)

5.Security: Role-Based Access Control (RBAC), Data Encryption, GDPR Compliance

Key Points:

1. **Technology Stack Used: python,vscode,gemini ai,**

Development Process:

1. Define Ethical Frameworks and Decision Rules

2. Collect and Preprocess Data
3. Develop AI Models for Decision-Making
4. 4. Implement Ethical Decision Analysis
5. User Interface (UI) for Interaction
6. Data Integrity & Security
7. Feedback Mechanism & Continuous Learning

8. **Challenges & Fixes:making of chartboard**

Phase-6: Functional & Performance Testing

Objective:

* AI Decision-Making & Analysis

* Ethical Framework Comparison

* Accuracy of AI Decisions

* Data Integrity & Security

Key Points:

Test Cases Executed: 1.User selects an ethical scenario

2. User inputs a free-text ethical decision
3. User chooses a multiple-choice response
4. AI explains an ethical decision based on different philosophical theories
5. AI provides risk scores and ethical impact analysis for a decision
6. AI is tested on edge cases
7. User requests AI to justify a past ethical decision
8. AI generates real-time ethical insights under high load
9. AI system provides accessible UI for visually impaired users

Bug Fixes & Improvements: ✅ Use Selenium or Cypress to automate UI interactions
✅ Validate API responses for scenario selection using **Postman or REST API testing tools**
✅ Check system logs for **error handling & loading performance**

Final Validation: User data and decisions are securely processed and stored,AI-generated recommendations must not favor any specific demographic, culture, or ideology,Users can browse, select, and engage with ethical dilemmas without errors

Deployment (if applicable): (Hosting details or final demo link)

Final Submission

1. **Project Report Based on the templates**
2. **Demo Video (3-5 Minutes)**
3. **GitHub/Code Repository Link**
4. **Presentation**
